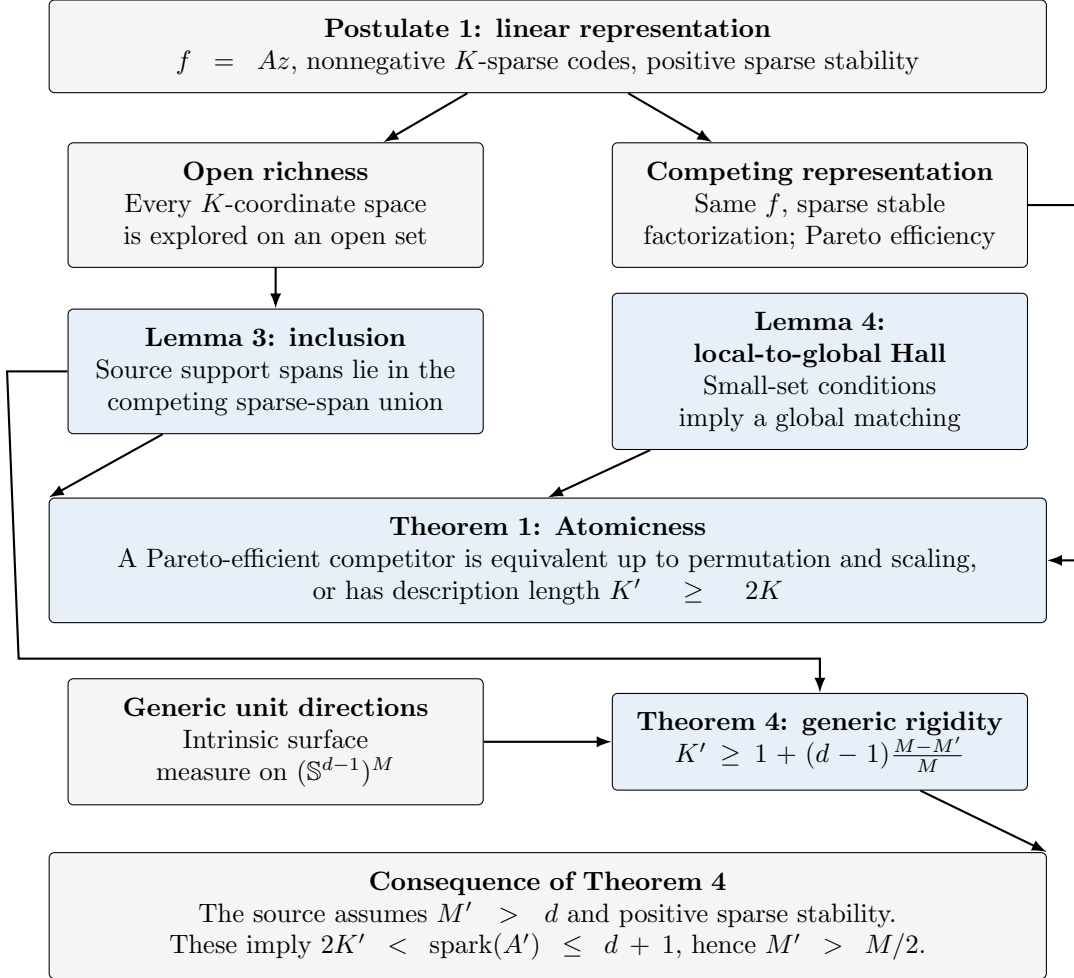


A Testable Theory of Atomic Features

Rigidity and generic dimension bounds

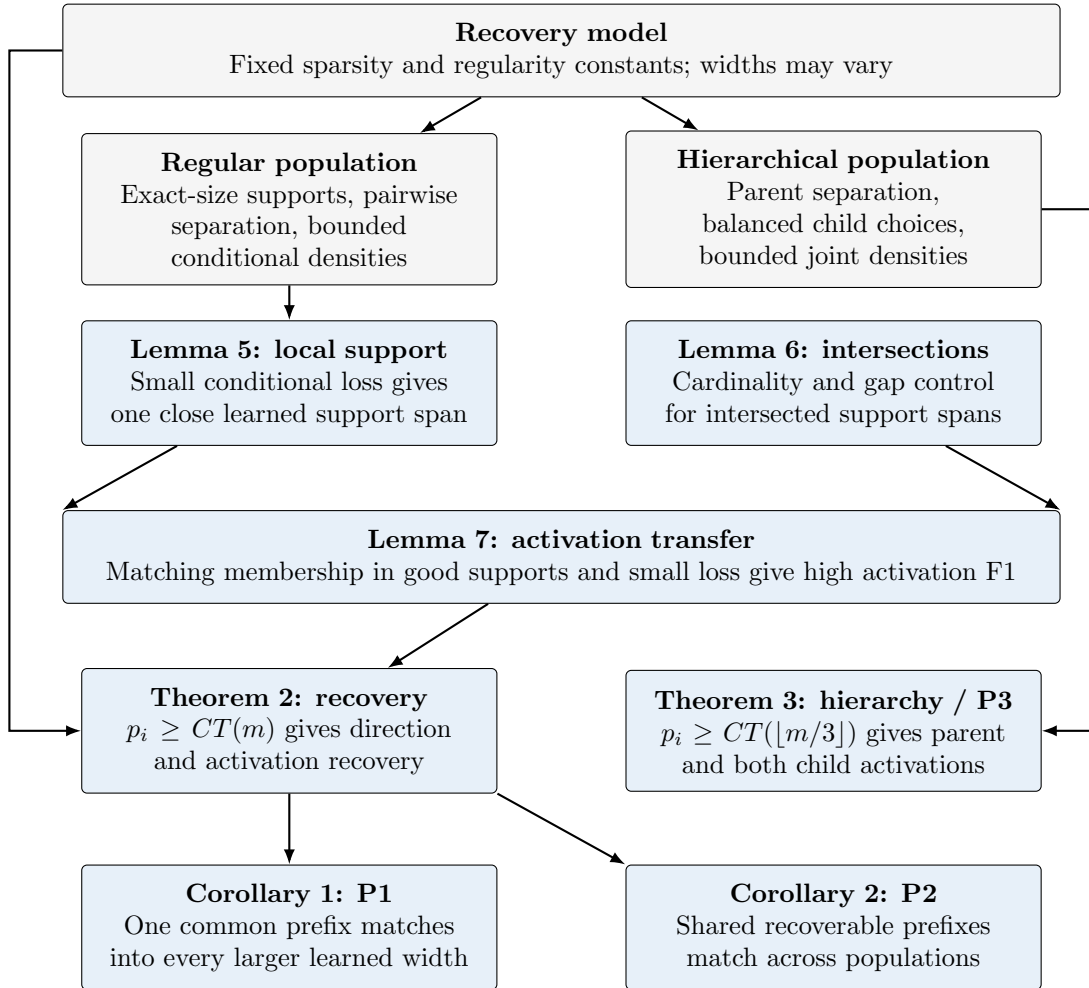
An arrow means that a definition or result supplies a mathematical input. The diagram shows paper-level statements; proof implementation details are omitted. Result numbers follow the September 25 manuscript.



Generic rigidity quantifies over overcomplete competitors, as stated in the source. Atomicness uses its own stated competing-dictionary class.

Recovery and its predictions

All recovery statements use normalized source and learned columns, positive sparse stability, nonnegative sparse codes, and a constant-factor bound on the actual population reconstruction loss. Targets have positive prevalence.



T is the discarded prevalence tail (the parent tail for hierarchy). No Zipf law or orthogonality is assumed. Hierarchy guarantees activation recovery; it does not assert ordinary direction recovery of every child. The support-pattern proof of Theorem 3 is described in the mathematical notes.